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Clinical Research Article

Polycystic Ovary Syndrome Underdiagnosis Patterns by Individual-level and Spatial Social Vulnerability Measures

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Abstract

Objective: To use electronic health records (EHR) data at Boston Medical Center (BMC) to identify individual-level and spatial predictors of missed diagnosis, among those who meet diagnostic criteria for polycystic ovary syndrome (PCOS).

Methods: The BMC Clinical Data Warehouse was used to source patients who presented between October 1, 2003, and September 30, 2015, for any of the following: androgen blood tests, hirsutism, evaluation of menstrual regularity, pelvic ultrasound for any reason, or PCOS. Algorithm PCOS cases were identified as those with International Classification of Diseases (ICD) codes for irregular menstruation and either an ICD code for hirsutism, elevated testosterone lab, or polycystic ovarian morphology as identified using natural language processing on pelvic ultrasounds. Logistic regression models were used to estimate odds ratios (ORs) of missed PCOS diagnosis by age, race/ethnicity, education, primary language, body mass index, insurance type, and social vulnerability index (SVI) score.

Results: In the 2003-2015 BMC-EHR PCOS at-risk cohort (n = 23 786), there were 1199 physician-diagnosed PCOS cases and 730 algorithm PCOS cases. In logistic regression models controlling for age, year, education, and SVI scores, Black/African American patients were more likely to have missed a PCOS diagnosis (OR = 1.69 [95% CI, 1.28, 2.24]) compared to non-Hispanic White patients, and relying on Medicaid or charity for insurance was associated with an increased odds of missed diagnosis when compared to private insurance (OR = 1.90 [95% CI, 1.47, 2.46], OR = 1.90 [95% CI, 1.41, 2.56], respectively). Higher SVI scores were associated with increased odds of missed diagnosis in univariate models.

Conclusion: We observed individual-level and spatial disparities within the PCOS diagnosis. Further research should explore drivers of disparities for earlier intervention.

Key Words: polycystic ovary syndrome, PCOS, disparities, clinical diagnosis

Polycystic ovary syndrome (PCOS), the most common endocrinology among reproductive-aged women, contributes a significant economic and health burden through its effects on reproductive, metabolic, and psychosocial health. The prevalence of PCOS is estimated to be between 6% and 12% in the United States (1). In 2020 alone, PCOS was associated with healthcare costs of ~$8 billion in the United States (2). PCOS is the leading cause of infertility in reproductive-aged women, and women with PCOS have an increased risk of developing metabolic syndrome and its associated comorbidities, including diabetes, cardiovascular disease, and obesity, throughout their life course (3-5).

Despite potential serious implications, many women who meet diagnostic criteria for PCOS remain undiagnosed or experience delays in diagnosis (6, 7). The Rotterdam Criteria for diagnosis, established in 2003, requires 2 out of the 3 cardinal features of PCOS: irregular menstruation, symptoms of androgen excess (ie, hirsutism) or lab tests for biochemical androgen excess, and the presence of ovarian cysts on transvaginal ultrasound (8, 9). According to current guidelines, if a provider suspects a patient has PCOS, they will inquire about relevant symptoms including menstrual cycle irregularity, hirsutism, acne, male-pattern hair loss, and weight gain (10, 11). If PCOS is suspected but not confirmed symptomatically, after ruling out other endocrinopathies that present with similar symptoms, a provider may order a test for biochemical hyperandrogenemia or a pelvic ultrasound to check for polycystic ovarian morphology to confirm diagnosis. After receiving a PCOS diagnosis, a patient ideally receives access to counseling around menstrual and uterine health, information about potential comorbidities and prevention strategies, and tailored fertility care during periods of attempted conception (12).

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Patients with PCOS may experience symptoms of metabolic syndromes 20 years earlier than non-PCOS patients, shifting disease burden and severity to the reproductive age (13). Fortunately, early therapeutic and nutritional interventions have been shown to lower the burden of diabetes and metabolic syndrome (14, 15). These interventions may be critically impactful in marginalized and disadvantaged groups. Historically, in the United States, across a variety of health outcomes, marginalized racial/ethnic and lower socioeconomic status (SES) groups are less likely to receive diagnoses and adequate medical care (16). Diabetes morbidity among reproductive-aged women disproportionately affects Black vs White women (17). Due to the potential mitigating effects of early intervention on poor health and quality-of-life outcomes, it is critical to understand the factors related to receiving a PCOS diagnosis and the timing of PCOS diagnosis.

Although studies have explored potential disparities in PCOS phenotypes, symptoms, and severity (18-23), little is known about how individual- and neighborhood-level race/ethnicity and SES factors predict receiving a diagnosis. Data from Boston Medical Center (BMC) electronic health records (EHR) is well-positioned to answer these questions due to its diverse patient population and availability of residential address history. As a safety net hospital, BMC serves a vulnerable patient population that may be missed in other hospitals and cohorts and therefore may better capture the true burden of disease in an US urban environment.

Here we aim to use hospital EHR data at BMC from 2003 to 2015 to identify predictors of not receiving a diagnosis among those who meet criteria for PCOS.

Materials and Methods

Study Population

A cohort of women at risk of PCOS was obtained from the BMC Clinical Data Warehouse including anyone who presented at BMC for primary care, endocrinology, family medicine, or obstetrics and gynecology between October 1, 2003, and September 30, 2015 for any of the following: androgen blood tests, hirsutism, evaluation of menstrual regularity, pelvic ultrasound for any reason, or PCOS. Patients diagnosed with endocrinopathies that must be ruled out for a PCOS diagnosis to be considered were excluded (Supplemental Table S1) (24). Physician-diagnosed cases were identified as females aged 18 to 45 years with an International Classification of Diseases (ICD)-9 code for PCOS (256.4). Algorithm PCOS cases were identified as those with ICD-9 codes for irregular menstruation (absence of menstruation, 626.0; scanty or infrequent menstruation, 626.1; irregular menstrual cycle, 626.4) and at least 1 of the following: ICD-9 code for hirsutism (704.1), elevated testosterone lab (total testosterone > 45 ng/dL), or polycystic ovarian morphology (PCOM) identified using natural language processing technique on pelvic ultrasound reports, as described previously (25). Diagnosis date was defined as the date the PCOS ICD-9 code was assigned for physician-diagnosed cases and as the date the first PCOS criteria was met for algorithm PCOS cases. A second algorithm PCOS definition, aligning with the Rotterdam criteria for diagnosis (8, 9) and additionally including patients with androgen excess and PCOM without medical record documentation of irregular menstruation, was used in sensitivity analyses. Although this group is covered by the Rotterdam criteria, they were not included in our primary study population as irregular menstruation is usually considered a key first criterion, and without full clinical evaluation it is unclear whether this group has true PCOS. The Boston University Medical Campus and Boston University School of Medicine Institutional Review Boards approved the protocol.

Individual-level Variables

Time-varying residential address and insurance type were updated at each visit. Demographic and socioeconomic variables including race/ethnicity, highest education achieved, primary language, and body mass index (BMI), were extracted from EHR. Insurance type included private, charity, Medicaid, or other (self-pay/uninsured, employer, other). Charity insurance is offered through BMC's Charity Care Program and grants financial assistance to all low-income, uninsured, or underinsured patients who are unable to pay for any portion of services (26). Race/ethnicity was categorized as Black/African American, non-Hispanic White, Hispanic/Latino, or other (combined due to small numbers: Asian, American Indian/Native American, Middle Eastern, and Native Hawaiian/Pacific Islander). Highest education was categorized as: some high school or less, graduated high school or General Education Development test (GED), at least some college/vocational/technical school, or other (declined, unavailable, or preregistration).

Variables extracted from medical records that capture belonging to a racial/ethnic group and SES (ie, insurance type, highest education) are not expected to have direct causal effects on the outcome of PCOS diagnosis. However, these variables were included as proxy measures of historical and contemporary structural and institutional racism as well as perpetuation of wealth inequality, which we hypothesize negatively impact the probability of receiving a diagnosis, among eligible patients with PCOS (27-29). BMI was included as a proxy measure for severity of metabolic symptoms. Previously published work by Howe et al recommend the use of causal diagrams to illustrate that race is a socially constructed, rather than biological, variable and that structural/institutional racism is causally related to health disparities (30). Based on their framework, we have developed a possible conceptual diagram relating the measured variables, unmeasured factors, PCOS development, and PCOS diagnosis (Fig. 1).

For patients with missing BMI at the date of diagnosis, values from the previous appointment were carried forward. Patients with missing BMI but available ICD codes for overweight, obesity, or class III obesity were assigned a BMI of 27.5, 35, and 40, respectively. The remaining missing BMI values were imputed using chained equation multiple imputation for all analyses. Imputation models for continuous BMI included all covariates and outcomes in the study. We generated 5 imputed datasets and used the Multivariate Imputation via Chained Equations R software to combine imputed datasets (31). The number of BMC visits by individual in the year and 2 years prior to and following diagnosis were calculated (based on visits where vitals were measured and recorded) as a measure of frequency of interaction with the medical system.

Area-level Variables

The Social Vulnerability Index (SVI), developed by the Agency for Toxic Substances and Disease Registry at the Centers for Disease Control and Prevention, is derived from census tract variables including percent below poverty, percent unemployed,

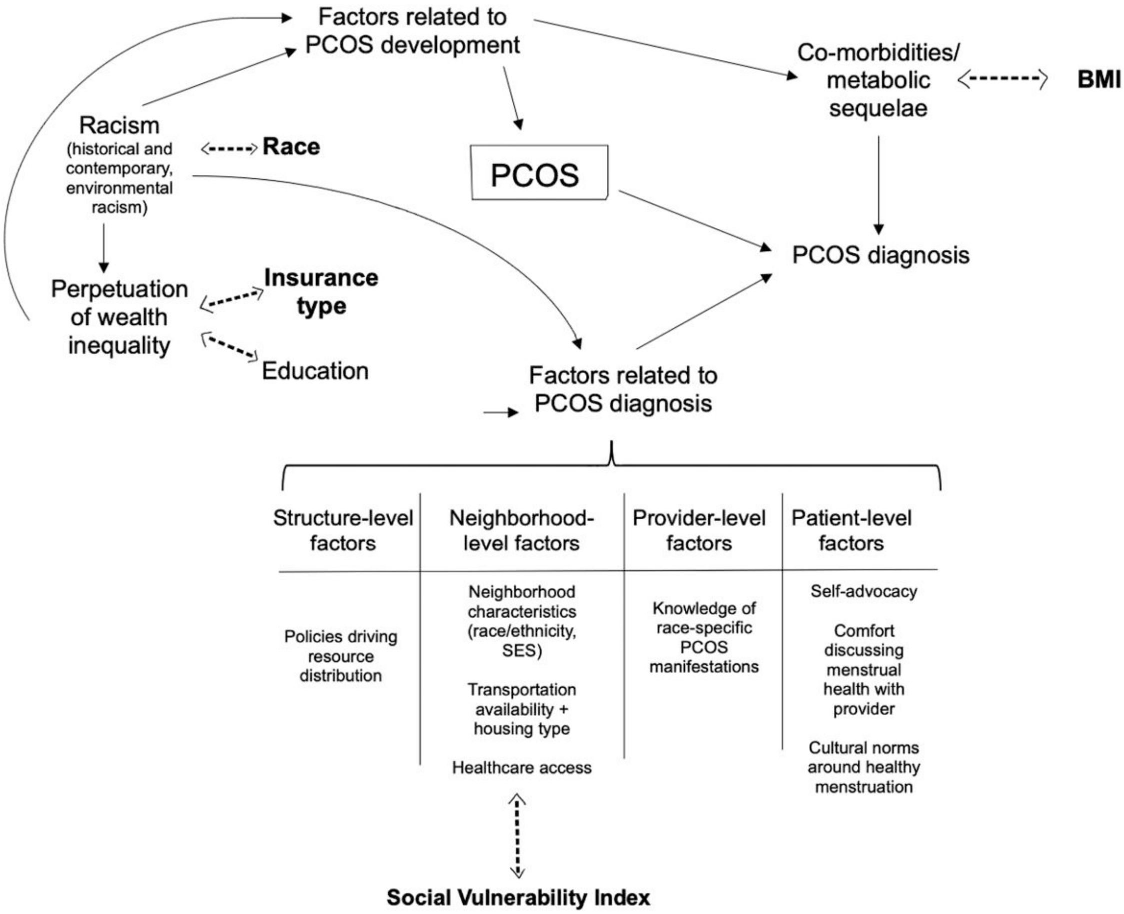


Figure 1. Proposed conceptual diagram for disparities in PCOS, PCOS comorbidities, and PCOS diagnoses.
Abbreviations: PCOS, polycystic ovary syndrome.

housing cost, percent without high school diploma, percent without insurance, percent English language proficiency, percent Hispanic/Latino, percent Black or African American, percent without vehicle, percent multiunit structures, etc. (32). SVI scores, calculated using data from the 2010 American Community Survey, were assigned by census tract to all residential addresses. SVI scores were calculated overall and for 4 subscores capturing socioeconomic, household composition/disability, minority status/language, and housing type/transportation. As census tracts are a contiguous unit containing between 1200 and 8000 residents and typically used to represent a neighborhood (33), SVI scores were included in our models as a proxy for neighborhood-level characteristics impacting the probability of receiving a PCOS diagnosis (Fig. 1).

Statistical Analyses

Time trends were plotted for ICD-9 PCOS cases and algorithm PCOS cases per year between 2003 and 2015. Patient demographic, socioeconomic, and SVI variables were summarized by mean (SD) and median (min, max) for continuous variables and count (percent) for categorical variables overall and by case group. *P*-values were calculated from two-sample *t*-test for numeric variables and chi-squared test of independence for categorical variables comparing ICD-9 and algorithm PCOS case populations, included in Table 1. Two-sample *t*-tests were used to compare the mean number of BMC visits in the years prior to and following diagnosis for ICD-9 and algorithm PCOS case groups. Logistic regression models were used to estimate the odds of missed PCOS diagnosis by demographic and socioeconomic factors available in EHR: age, race/ethnicity, highest education obtained, primary language, BMI, and insurance type. Total SVI score was included in fully adjusted models to adjust for confounding not captured by individual-level factors. Race/ethnicity, education, language, BMI, and insurance type were modeled as categorical variables. Age and calendar year were modeled using cubic splines to account for nonlinearity and nonnormal distributions. SVI scores were transformed to percentage points of the distribution represented in our study population. Univariate models and fully adjusted models were fit for each covariate. In a secondary analysis, we examined multiplicative interaction terms between race/ethnicity and insurance type. Total SVI and individual SVI theme subscores were evaluated in univariate logistic models and then were each added to fully adjusted models individually, due to high correlation between subscores.

Results

The cohort was 47% Black/African American, 25% non-Hispanic White, and 15% Hispanic/Latino, and 56% of women were on charity or Medicaid insurance (Table 1). A total of 1199 physician-diagnosed PCOS cases and 730 algorithm PCOS cases from 2003 to 2015 were identified in the

Table 1. Summary demographics and social vulnerability index scores for BMC-EHR PCOS cohort between 2003 and 2015 by PCOS diagnosis (n = 1929)

	PCOS ICD-9[a] (n = 1199)	Algorithm PCOS[a] (n = 730)	Overall (n = 1929)	
Age				
Mean (SD)	26.8 (5.95)	28.2 (6.92)	27.4 (6.37)	<.001[a]
Median [Min, Max]	26.0 [18.0, 45.0]	27.0 [18.0, 44.0]	26.0 [18.0, 45.0]	
Race/ethnicity, n (%)				
Non-Hispanic White	334 (27.9)	138 (18.9)	472 (24.5)	<.001[a]
Black/African American	495 (41.3)	402 (55.1)	897 (46.5)	
Hispanic/Latino	193 (16.1)	98 (13.4)	291 (15.1)	
Other	177 (14.8)	92 (12.6)	269 (13.9)	
Highest education, n (%)				
Some high school or less	277 (23.1)	204 (27.9)	481 (24.9)	.001[a]
Graduated high school or GED	299 (24.9)	206 (28.2)	505 (26.2)	
At least some college/vo-tech program	474 (39.5)	261 (35.8)	735 (38.1)	
Other	149 (12.4)	59 (8.1)	208 (10.8)	
Language, n (%)				
English	1000 (83.4)	557 (76.3)	1557 (80.7)	<.001[a]
Other	199 (16.6)	173 (23.7)	372 (19.3)	
BMI, n (%)				
Under/normal weight	172 (14.3)	137 (18.8)	309 (16.0)	<.001[a]
Overweight	211 (17.6)	63 (8.6)	274 (14.2)	
Obese	424 (35.4)	195 (26.7)	619 (32.1)	
Class III obese	205 (17.1)	125 (17.1)	330 (17.1)	
Missing	187 (15.6)	210 (28.8)	397 (20.6)	
Insurance type, n (%)				
Private	501 (41.8)	206 (28.2)	707 (36.7)	<.001[a]
Charity	210 (17.5)	181 (24.8)	391 (20.3)	
Medicaid	399 (33.3)	296 (40.5)	695 (36.0)	
Other	89 (7.4)	47 (6.4)	136 (7.1)	
Year of diagnosis				
Mean (SD)	2010 (3.58)	2010 (3.46)	2010 (3.55)	<.001[a]
Median [Min, Max]	2010 [2003, 2015]	2010 [2003, 2015]	2010 [2003, 2015]	
Number of visits in 1 year prior to diagnosis				
Mean (SD)	2.79 (2.84)	2.57 (3.35)	2.71 (3.05)	.14[a]
Median [Min, Max]	2.00 [0, 27.0]	1.00 [0, 22.0]	2.00 [0, 27.0]	
Missing (%)	17 (1.4)	6 (.8)	23 (1.2)	
Number of visits in 1 year after diagnosis				
Mean (SD)	3.78 (4.15)	3.88 (4.57)	3.82 (4.31)	.65[a]
Median [Min, Max]	3.00 [0, 26.0]	2.00 [0, 38.0]	2.00 [0, 38.0]	
Missing (%)	17 (1.4)	6 (.8)	23 (1.2)	
Standardized Social Vulnerability Index Score				
Mean (SD)	6.58 (1.71)	6.91 (1.65)	6.70 (1.70)	<.001[a]
Median [Min, Max]	6.69 [1.54, 10.0]	7.16 [2.78, 10.0]	6.89 [1.54, 10.0]	
Standardized Social Vulnerability Index Score: Socioeconomic Factor				
Mean (SD)	5.74 (2.48)	6.16 (2.40)	5.90 (2.45)	<.001[a]
Median [Min, Max]	5.88 [.114, 10.0]	6.31 [.568, 10.0]	6.03 [.114, 10.0]	
Standardized Social Vulnerability Index Score: Household Composition/Disability Factor				
Mean (SD)	6.04 (2.73)	6.35 (2.61)	6.16 (2.68)	.014[a]
Median [Min, Max]	6.43 [.044, 10.0]	6.69 [.044, 10.0]	6.51 [.044, 10.0]	
Standardized Social Vulnerability Index Score: Minority Status/Language Factor				
Mean (SD)	7.49 (2.06)	7.85 (1.82)	7.62 (1.98)	<.001[a]
Median [Min, Max]	8.12 [.238, 9.97]	8.36 [.534, 10.0]	8.21 [.238, 10.0]	

(*continued*)

Table 1. Continued

	PCOS ICD-9[a] (n = 1199)	Algorithm PCOS[a] (n = 730)	Overall (n = 1929)	
Standardized Social Vulnerability Index Score: Housing Type/Transportation Factor				
Mean (SD)	6.13 (1.39)	6.33 (1.28)	6.21 (1.35)	.001[a]
Median [Min, Max]	6.29 [.195, 10.0]	6.51 [1.89, 10.0]	6.43 [.195, 10.0]	

Abbreviations: BMC-EHR, Boston Medical Center electronic health records; BMI, body mass index; ICD-9, International Classification of Diseases, ninth revision; PCOS, polycystic ovary syndrome.
[a]*P*-values calculated from two-sample *t*-test for numeric variables and chi-squared test of independence for categorical variables.

BMC-EHR cohort (mean age ± SD, 27.4 ± 6.4) from a source population of 23 786. The algorithm PCOS group included 248 patients who qualified due to irregular menstruation and androgen excess, 436 patients with irregular menstruation and PCOM, and 46 patients who met all 3 criteria. Over the study period, the number of patients per year assigned PCOS ICD-9 codes increased while the number who met the criteria for algorithm PCOS remained relatively stable (Fig. 2). Both ICD-9 and algorithm PCOS cases appear to drop in 2015; however, our study ended in September 2015 when BMC began adopting ICD-10 codes. We observed no meaningful differences in number of visits to BMC in the year prior to ($P = .14$) or the year following ($P = .65$) diagnosis comparing the physician-diagnosed PCOS and algorithm PCOS groups.

Our primary model controlled for age, race/ethnicity, education, primary language, BMI, insurance type, and overall SVI score. Age and calendar year were included as cubic spline terms due to nonlinearity. Odds of missed PCOS diagnosis were higher in younger and older patients. Odds of missed diagnosis decreased over the study period but more dramatically in later years (2013-2015). Being Black/African American vs non-Hispanic White was associated with an OR for missed PCOS diagnosis of 1.69 (95% CI, 1.28, 2.24), and having Medicaid or charity vs private insurance was associated with an OR for missed diagnosis of 1.90 (95% CI, 1.47, 2.46) and 1.90 (95% CI, 1.41, 2.56), respectively (Fig. 3). For non-English speakers compared to English speakers, the OR for missed diagnosis was 1.34 (95% CI, 1.03, 1.75).

Patients with higher BMI had lower odds of missed PCOS diagnosis when compared to patients with lower BMI. Odds of missed PCOS diagnosis decreased over the study period. Education category was not associated with odds of missed diagnosis in fully adjusted models. The nonsignificant protective effect observed for the "other" education group was primarily driven by patients whose education was marked "unavailable" on EHR. Effect estimates were robust to adjustment for SVI score. Sensitivity analyses using the algorithm PCOS definition including patients with androgen excess and PCOM but no record of irregular menstruation were similar to those presented for the primary cohort (results not shown). In a secondary analysis, we observed a statistically significant interaction between charity insurance type and race/ethnicity, suggesting the association between charity insurance and increased odds of missed PCOS diagnosis is higher among non-Hispanic White patients than Black/African American and Hispanic/Latino patients (Supplemental Table S2) (24).

Total SVI score and each SVI subscore (socioeconomic, household composition, minority status/language, and housing type/transportation) were associated with increased odds of missed PCOS diagnosis in univariate models (Fig. 4). The SVI housing type/transportation factor was the only subscore independently associated with missed diagnosis (OR = 1.10 [95% CI, 1.02, 1.18] per 10% increase in SVI score) in fully adjusted models (Fig. 4).

Discussion

In our cohort of women at risk for PCOS at a Boston safety net hospital, we observed a large proportion of undiagnosed PCOS cases. The probability of missed diagnosis was higher among Black/African American vs non-Hispanic White women, among women on charity or Medicaid vs private insurance, among non-English vs English speakers, and among women with lower vs higher BMI. The probability of receiving a diagnosis, among women who met the criteria, increased over the study period from 2003 to 2015. Increased census-tract vulnerability score for socioeconomic, household composition, minority status/language, and housing type/transportation factors were also associated with a higher probability of missed diagnosis.

Two prior studies have examined differences between women with PCOS diagnoses and women who met PCOS criteria but did not receive a diagnosis. Ezeh et al compared PCOS cases at an outpatient clinic and a group of undiagnosed PCOS cases identified in a cohort undergoing preemployment physical exams between 1995 and 1999 and showed that women who were non-Hispanic White, had high BMI, and had high hirsutism scores were more likely to receive a diagnosis of PCOS vs unrecognized PCOS (34). The second study by Kim et al evaluated self-reported PCOS diagnoses and unrecognized PCOS in the Coronary Artery Risk Development in Young Adults (CARDIA) study, a prospective longitudinal cohort of US Black and White adults followed from 1985 to 2001 (35). Patients who self-reported PCOS diagnoses had higher BMI and higher prevalence of diabetes. There were no significant race differences between the diagnosed PCOS, unrecognized PCOS, and non-PCOS groups in the CARDIA cohort. Both studies differed from ours in that they defined PCOS using the 1990 National Institute of Health criteria, which require irregular menstruation and androgen excess for diagnosis (36), and were not able to capture suspected PCOS cases who currently qualify for diagnosis based on polycystic ovaries from ultrasound scans. Our finding that patients who are non-Hispanic White are more likely to receive PCOS diagnoses compared to eligible patients who are Black/African American is consistent with Ezeh et al. Consistent with the prior studies showing diagnosed PCOS cases presented with more severe phenotypes/metabolic sequelae compared to undiagnosed cases, we found patients meeting PCOS

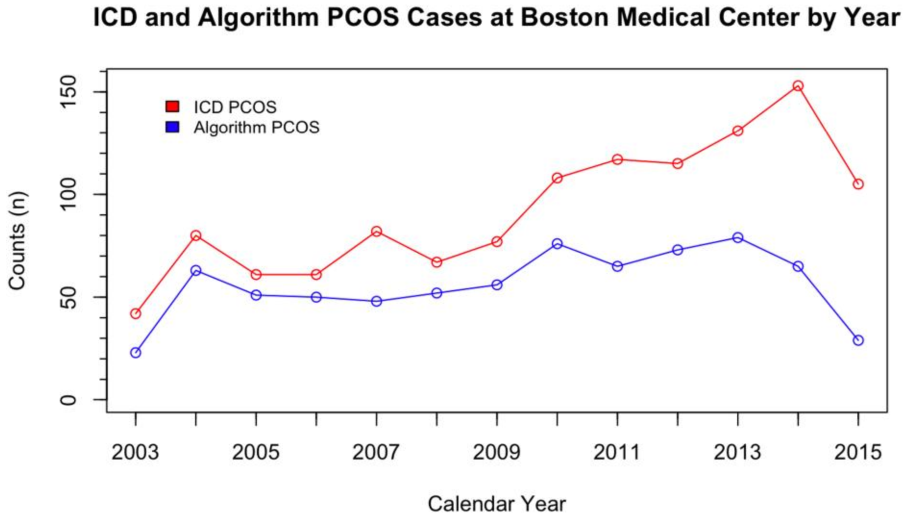
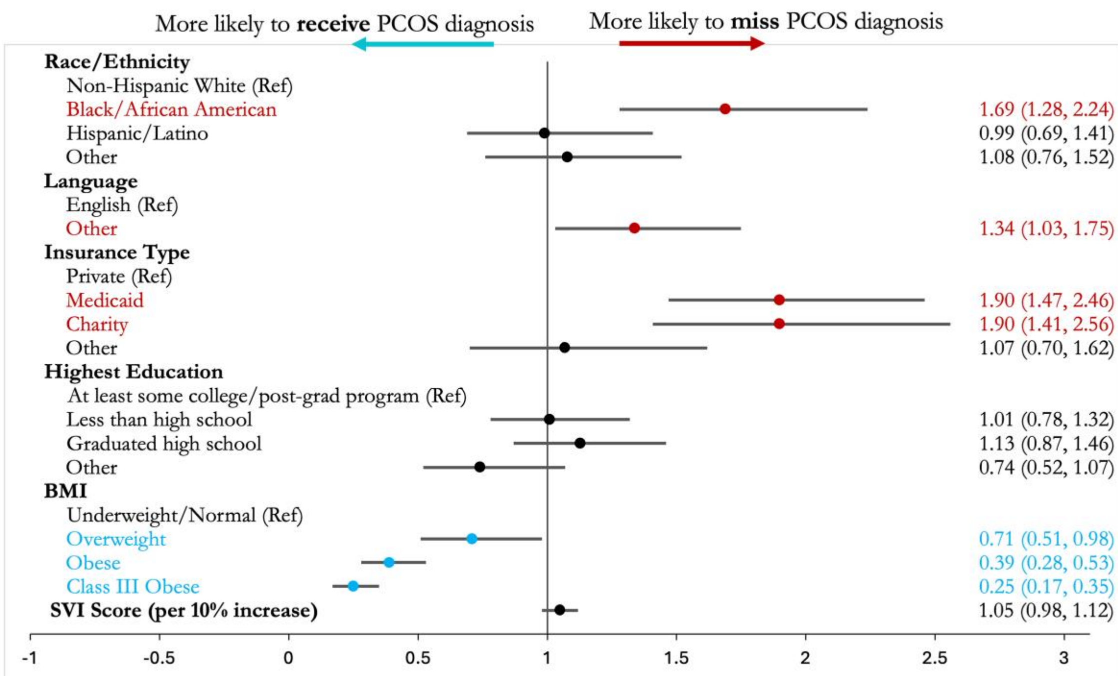


Figure 2. Time trends for number of International Classification of Diseases, ninth revision, PCOS and algorithm PCOS cases per year at the Boston Medical Center between 2003 and 2025 (n = 1929).

Abbreviations: PCOS, polycystic ovary syndrome.



IQR = interquartile range; GED = general equivalency diploma; dx = diagnosis; BMI = body mass index; SVI = Social Vulnerability Index
IQR for age = 10 years; IQR for year dx = 6 years
Statistically significant estimates ($p < 0.05$) are highlighted in red if they are > 1 (more likely to miss diagnosis) and teal if they are < 1 (more likely to receive diagnosis).

Figure 3. Odds ratios for the probability of missed diagnoses by age, race/ethnicity, education, and insurance type among those who meet the criteria for polycystic ovary syndrome diagnosis (n = 1929) for women aged 18 to 45 attending Boston Medical Center between 2003 and 2015.

criteria with lower BMI were less likely to receive diagnoses. Our study is the first to examine both individual-level and spatial vulnerability and the first to compare diagnosed and undiagnosed PCOS cases in a single hospital-based cohort.

One potential hypothesis for a driver of the observed disparities in PCOS diagnoses by markers of race/ethnicity and social vulnerability is the well-documented differences in access to and utilization of healthcare resources among lower SES and minority race/ethnicity groups (37). Reasons for these disparities include low income and lack of insurance coverage (38, 39), difficulty accessing a consistent location for care (37, 40), language barriers (41), and availability of transit/travel burden (42). In the CARDIA study, the authors observed no differences in self-reported access to medical care in adulthood across undiagnosed and diagnosed PCOS groups, although this observation could be attributable to the method and timing of social determinants of health (SDoH) assessment (35). In our analysis we attempted to capture individual-level access to and interaction with the medical system by comparing the number of BMC visits in the years prior to and following PCOS diagnosis, and we did not observe meaningful differences across groups. Number of visits is related to

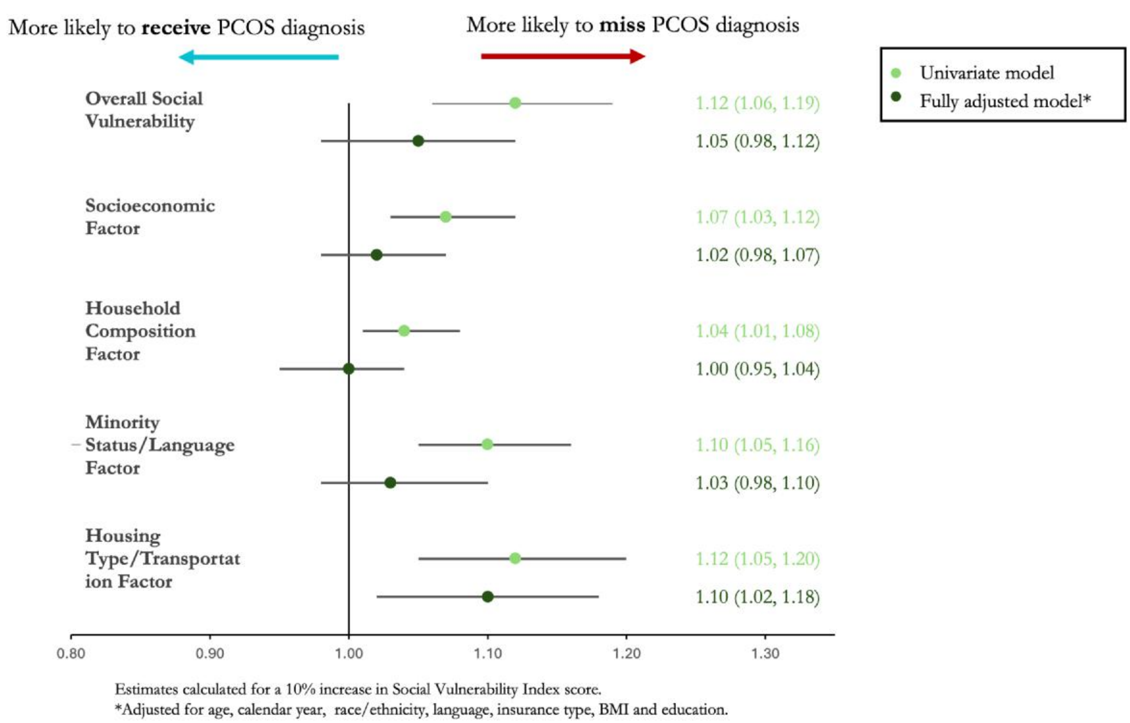


Figure 4. Odds ratios for the probability of missed diagnoses by Social Vulnerability Index overall and factor scores among those who meet the criteria for polycystic ovary syndrome diagnosis (n = 1929) for women aged 18 to 45 attending Boston Medical Center between 2003 and 2015.

both access to care and health status, 2 factors that are difficult to disentangle using medical record data. However, we did observe a higher probability of missed diagnosis among women using public vs private insurance, a marker of SES and healthcare access.

Disparities in access to care have also been studied using neighborhood deprivation or vulnerability scores calculated from census-tract data (43). In Brazil, high SVI scores identified geographical regions with low basic healthcare access, suggesting that limits in access may be related to SDoH (44). We employed SVI scores to attempt to capture hospital access and SDoH at the spatial level. We observed that social vulnerability scores indicating greater vulnerability for SES, household composition, minority status/language, and housing type/transportation were associated with a higher probability of missed PCOS diagnosis. One previous study comparing PCOS-diagnosed cases and controls in a hospital-based cohort found that PCOS diagnosis rates were lowest in patients with high area-level deprivation index scores (45). Although this study was unable to isolate undiagnosed PCOS cases, the authors acknowledged that neighborhood disadvantage may play a role in the high rate of undiagnosed patients. Increased utilization of telehealth post-COVID-19 pandemic renews the importance of a focus on access to care through digital platforms. An analysis of electronic records in the Northeast United States during the pandemic showed that patients taking advantage of virtual vs in person care were more likely to be older and English-proficient, and audio only visits were an important option for the most vulnerable patients (46). In contrast, an analysis of patients seeking care for gynecological cancer during the pandemic found that virtual visits were utilized by patients of all ages and cancer types, highlighting the potential of telemedicine to improve access to care across a range of social vulnerabilities (47). Virtual healthcare may improve access for groups with limited time or transportation but also could widen the gap if care is not taken to reach historically underserved communities. Expanding analyses on diagnosis disparities by social vulnerability through the current day will help evaluate the effect of the pandemic on access to gynecological care and identify areas for intervention.

Another potential explanation for observed disparities in PCOS includes the lack of clarity on racial/ethnic-specific manifestations of PCOS-related symptoms. For example, acanthosis nigricans, a dermatologic manifestation of hyperandrogenism and insulin resistance, is much more common in people with Black and South Asian ancestry than Whites; however, hirsutism and severe acne remain the primary indicators of clinical androgen excess for assigning PCOS diagnoses (48-50). Clarifying guidelines and ensuring provider education of racial/ethnic-specific criteria cutoffs and common manifestations of PCOS will be important for timely PCOS diagnoses for patients of all race/ethnicities. Previous studies have shown differences in PCOS phenotype, severity, and comorbidities by race/ethnicity. In some studies, fasting insulin levels, insulin resistance, and metabolic syndrome prevalence were higher among Hispanic White women with PCOS (51-54), while in other studies they were higher among Black women with PCOS (54-56), and in others there were no significant differences across racial/ethnic groups (19, 57-59). However, based on previous literature, we expect women with more severe phenotypes to be more likely to receive diagnoses (34, 35). Inconsistent observations of disparities in PCOS phenotype and severity may be related to underdiagnosis; if disadvantaged groups experience barriers to diagnosis, only the most severe PCOS patients in these groups will receive diagnoses. In our analysis we included adjustment for BMI, which only strengthened effect estimates for race, insurance type, and language, suggesting that the disparities observed are not explained by increased risk of syndrome severity among disadvantaged groups. Our results emphasize the importance of using unselected population-based cohorts or cohorts including undiagnosed algorithm PCOS cases to disentangle racial/ethnic-specific manifestations of PCOS, underdiagnosis by socially vulnerable groups, and differences in PCOS prevalence or PCOS symptom severity for people belonging to specific racial/ethnic groups.

Although the use of EHR data allowed for utilization of a new method for assigning algorithm PCOS among eligible patients in a hospital cohort, our study has limitations. Because we examined the probability of receiving a diagnosis among women who met the criteria for PCOS in a hospital cohort, we were unable to estimate the prevalence of diagnosed and unrecognized PCOS or to compare demographics and predictors to an unselected control population. Additionally, EHR data is always subject to informative presence bias as patients with poorer health are more likely to be seen frequently in the hospital (60). However, our results are mostly consistent with existing studies in unselected populations, and it remains important to examine diagnosis disparities at the clinic level among patients currently eligible for diagnoses. We were only able to estimate associations for well-represented race/ethnicity groups and insurance types, and "other" groups comprise a heterogenous mix of participants.

Our study was also subject to covariate missingness in medical records. Due to the significant missingness within the BMI variable, we assumed missing BMI was missing at random conditional on observed medical record data and imputed missing values for ~20% of our cohort, in which case multiple imputation of missing BMI would lead to unbiased estimates. Effect estimates were similar when the study population was restricted to those with complete BMI data (results not shown). Our results may not be generalizable outside of the BMC cohort. BMC is a safety net hospital that offers exceptional language interpreter services. In 2022, BMC was ranked the number 1 most socially responsible hospital in Massachusetts and the number 1 most racially inclusive hospital in the United States by the Lown Institute (61, 62). Healthcare systems that are further underresourced may experience even wider disparities. We were limited in our ability to evaluate additional potential confounding factors (ie, lifestyle habits, family and personal medical history) and physician's characteristics (ie, age, race/ethnicity, specialty) due to data availability in medical records. Finally, we were limited in that we were unable to definitively identify the underlying drivers of observed disparities. Multilevel mediators may explain disparities: at the spatial level through transportation and healthcare access, at the provider level through knowledge of racial/ethnic-specific PCOS manifestations, and at the patient level through language proficiency and ability to self-advocate (Fig. 2). Our study generates hypotheses for future studies, which should utilize advanced statistical methodology to estimate the effects of multiple mediators on PCOS diagnosis disparities (30, 63).

Our study is the first to examine PCOS and unrecognized PCOS in a hospital population. We are also the first to show that insurance type, language, and spatial vulnerability predict the probability of a missed diagnosis. Using EHR, we were able to identify cases with records of irregular menstruation, records of clinical and biochemical androgen excess, and ultrasound evidence of PCOM. Recent guidelines from the American College of Obstetrics and Gynecologists highlight the presence of both implicit bias leading to disparities in the way health care is delivered and structural racism leading to downstream effects on social determinants of health (28, 64). The first step to closing gaps in reproductive health requires recognizing the existing disparities (16). The second step requires attention to racial/ethnic-specific diagnosis criteria, provider bias training, and systemic factors in place that affect healthcare access. We observed improved recognition and assignment of PCOS diagnoses over the study period, which may be partially attributable to efforts to clarify diagnostic criteria (6, 7). Ensuring vulnerable patients receive timely PCOS diagnoses will increase their access to information and support, reduce comorbidity risk, and improve quality of life. Cohorts developed to study PCOS should include algorithm PCOS cases to attempt to capture the large unrecognized patient population. Future papers on PCOS diagnoses should explore drivers of disparities to inform areas for earlier intervention.

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Disclosures

The authors declare no relevant or material conflicts of interest that relate to the research described in this paper.

Data Availability

Restrictions apply to the availability of data analyzed during this study to preserve patient confidentiality. The corresponding author will on request detail the restrictions and any conditions under which access to some data may be provided.

References

1. Centers for Disease Control and Prevention. *PCOS (Polycystic Ovary Syndrome) and Diabetes*. Centers for Disease Control and Prevention; Published March 24, 2020. Accessed October 3, 2023. https://www.cdc.gov/diabetes/risk-factors/pcos-polycystic-ovary-syndrome.html
2. Riestenberg C, Jagasia A, Markovic D, Buyalos RP, Azziz R. Health care-related economic burden of polycystic ovary syndrome in the United States: pregnancy-related and long-term health consequences. *J Clin Endocrinol Metab*. 2022;107(2):575-585.
3. Joham A, Palomba S, Hart R. Polycystic ovary syndrome, obesity, and pregnancy. *Semin Reprod Med*. 2016;34(02):093-101.
4. Dumesic DA, Oberfield SE, Stener-Victorin E, Marshall JC, Laven JS, Legro RS. Scientific statement on the diagnostic criteria, epidemiology, pathophysiology, and molecular genetics of polycystic ovary syndrome. *Endocr Rev*. 2015;36(5):487-525.

5. Boomsma CM, Eijkemans MJC, Hughes EG, Visser GHA, Fauser BCJM, Macklon NS. A meta-analysis of pregnancy outcomes in women with polycystic ovary syndrome. *Hum Reprod Update*. 2006;12(6):673-683.
6. Witchel SF, Teede HJ, Pena AS. Curtailing PCOS. *Pediatr Res*. 2020;87(2):353-361.
7. Hoeger KM, Dokras A, Piltonen T. Update on PCOS: consequences, challenges, and guiding treatment. *J Clin Endocrinol Metab*. 2021;106(3):e1071-e1083.
8. Rotterdam ESHRE/ASRM-Sponsored PCOS Consensus Workshop Group. Revised 2003 consensus on diagnostic criteria and long-term health risks related to polycystic ovary syndrome (PCOS). *Hum Reprod*. 2004;19(1):41-47.
9. Teede HJ, Tay CT, Laven JJE, *et al*. Recommendations from the 2023 international evidence-based guideline for the assessment and management of polycystic ovary syndrome. *Eur J Endocrinol*. 2023;189(2):G43-G64.
10. Goodman NF, Cobin RH, Futterweit W, Glueck JS, Legro RS, Carmina E. American Association of Clinical Endocrinologists, American College of Endocrinology, and androgen excess and PCOS society disease state clinical review: guide to the best practices in the evaluation and treatment of polycystic ovary syndrome—part 1. *Endocr Pract*. 2015;21(11):1291-1300.
11. Christ JP, Cedars MI. Current guidelines for diagnosing PCOS. *Diagnostics*. 2023;13(6):1113.
12. Al Wattar BH, Fisher M, Bevington L, *et al*. Clinical practice guidelines on the diagnosis and management of polycystic ovary syndrome: a systematic review and quality assessment study. *J Clin Endocrinol Metab*. 2021;106(8):2436-2446.
13. Apridonidze T, Essah PA, Iuorno MJ, Nestler JE. Prevalence and characteristics of the metabolic syndrome in women with polycystic ovary syndrome. *J Clin Endocrinol Metab*. 2005;90(4):1929-1935.
14. Bagchi D, Sreejayan N. *Nutritional and Therapeutic Interventions for Diabetes and Metabolic Syndrome*. 1st ed. Elsevier Academic Press; 2012.
15. Hsu CN, Hou CY, Hsu WH, Tain YL. Early-life origins of metabolic syndrome: mechanisms and preventive aspects. *Int J Mol Sci*. 2021;22(21):11872.
16. Geiger JH. Racial and ethnic disparities in diagnosis and treatment: a review of the evidence and a consideration of causes. In: *Unequal Treatment: Confronting Racial and Ethnic Disparities in Health Care*. National Academies Press (US); 2003:18-124. Doi: 10.17226/12875
17. Britton LE, Hussey JM, Crandell JL, Berry DC, Brooks JL, Bryant AG. Racial/ethnic disparities in diabetes diagnosis and glycemic control among women of reproductive age. *J Women's Health (Larchmont)*. 2018;27(10):1271-1277.
18. Merkin SS, Azziz R, Seeman T, *et al*. Socioeconomic status and polycystic ovary syndrome. *J Women's Health (Larchmont)*. 2011;20(3):413-419.
19. Ladson G, Dodson WC, Sweet SD, *et al*. Racial influence on the polycystic ovary syndrome phenotype: a black and white case-control study. *Fertil Steril*. 2011;96(1):224-229.e2.
20. Merkin SS, Phy JL, Sites CK, Yang D. Environmental determinants of polycystic ovary syndrome. *Fertil Steril*. 2016;106(1):16-24.
21. Rubin KH, Andersen MS, Abrahamsen B, Glintborg D. Socioeconomic status in Danish women with polycystic ovary syndrome: a register-based cohort study. *Acta Obstet Gynecol Scand*. 2019;98(4):440-450.
22. Ashraf S, Aslam R, Bashir I, Majeed I, Jamshaid M. Environmental determinants and PCOS symptoms severity: a cross-sectional study. *Health Care Women Int*. 2022;43(1-3):98-113.
23. Kazemi M, Kim JY, Parry SA, Azziz R, Lujan ME. Black women have a worse cardio-metabolic risk profile compared to white women with polycystic ovary syndrome in the United States: a systematic review and meta-analysis. *J Endocr Soc*. 2021;5(Supplement_1): A283-A284.
24. Silva E. Supplementary Material for "Polycystic Ovary Syndrome Underdiagnosis Patterns by Individual-level and Spatial Social Vulnerability Measures". *Zenodo*. 2024. https://doi.org/10.5281/zenodo.13999374
25. Cheng JJ, Mahalingaiah S. Data mining polycystic ovary morphology in electronic medical record ultrasound reports. *Fertil Res and Pract*. 2019;5(1):13.
26. Boston Medical Center. *Patient Financial Assistance*. Accessed November 6, 2023. https://www.bmc.org/patient-financial-assistance
27. Sutton MY, Anachebe NF, Lee R, Skanes H. Racial and ethnic disparities in reproductive health services and outcomes, 2020. *Obstet Gynecol*. 2021;137(2):225-233.
28. Gillispie-Bell V. The contrast of color: why the black community continues to suffer health disparities. *Obstet Gynecol*. 2021;137(2): 220-224.
29. Griffin A, Reed D. Unlearning, undoing, and uplifting: an organized approach to ending health disparities affecting black women. *Obstet Gynecol*. 2021;137(2):218-219.
30. Howe CJ, Bailey ZD, Raifman JR, Jackson JW. Recommendations for using causal diagrams to study racial health disparities. *Am J Epidemiol*. 2022;191(12):1981-1989.
31. Zhang Z. Multiple imputation with multivariate imputation by chained equation (MICE) package. *Ann Transl Med*. 2016;4(2):30.
32. Flanagan BE, Gregory EW, Hallisey EJ, Heitgerd JL, Lewis B. A social vulnerability Index for disaster management. *J Homel Secur Emerg Manag*. 2011;8(1):0000102202154773551792.
33. US Census Bureau. *Glossary*. Census.gov. Accessed March 19, 2024. https://www.census.gov/programs-surveys/geography/about/glossary.html
34. Ezeh U, Yildiz BO, Azziz R. Referral bias in defining the phenotype and prevalence of obesity in polycystic ovary syndrome. *J Clin Endocrinol Metab*. 2013;98(6):E1088-E1096.
35. Kim C, Schreiner PJ, Siscovick D, *et al*. Factors associated with self-report of polycystic ovary syndrome in the coronary artery risk development in young adults study (CARDIA). *BMC Women's Health*. 2023;23(1):248.
36. Azziz R. Diagnostic criteria for polycystic ovary syndrome: a reappraisal. *Fertil Steril*. 2005;83(5):1343-1346.
37. National Academies of Sciences, Engineering, and Medicine; Health and Medicine Division; Board on Health Care Services; Committee on Health Care Utilization and Adults with Disabilities. *Health-Care Utilization as a Proxy in Disability Determination*. National Academies Press (US); 2018. Doi: 10.17226/24969
38. Office of the Assistant Secretary for Planning and Evaluation; US Department of Health and Human Services, Office of Minority Health. *HHS Action Plan to Reduce Racial and Ethnic Health Disparities: Implementation Progress Report 2011-2014*. 2015. Accessed October 17, 2024. https://aspe.hhs.gov/reports/hhs-action-plan-reduce-racial-ethnic-health-disparities-implementation-progress-report-2011-2014-0
39. Barnett Jessica C, Vornovitsky Marina S, US Census Bureau. *Health Insurance Coverage in the United States: 2015*. 2016. https://www.census.gov/content/dam/Census/library/publications/2016/demo/p60-257.pdf
40. US Department of Health & Human Services. *2009 National Healthcare Quality and Disparities Reports | AHRQ Archive*. 2010. Accessed October 17, 2024. https://archive.ahrq.gov/research/findings/nhqrdr/nhqrdr09/qrdr09.html
41. Timmins CL. The impact of language barriers on the health care of Latinos in the United States: a review of the literature and guidelines for practice. *J Midwifery Womens Health*. 2002;47(2):80-96.
42. Syed ST, Gerber BS, Sharp LK. Traveling towards disease: transportation barriers to health care access. *J Community Health*. 2013;38(5): 976-993.
43. Mah JC, Penwarden JL, Pott H, Theou O, Andrew MK. Social vulnerability indices: a scoping review. *BMC Public Health*. 2023;23(1): 1253.
44. de Souza KOC, Fracolli LA, Ribeiro CJN, de Menezes AF, Silva GM, Santos ADD. Quality of basic health care and social

vulnerability: a spatial analysis. *Rev Esc Enferm USP*. 2021;55: e20200407.

45. Actkins KV, Aldrich M, Davis LK. A polycystic ovary syndrome diagnosis is a marker of socioeconomic advantage. *Fertil Steril*. 2021;116(3):e405.
46. Zachrison KS, Yan Z, Sequist T, *et al.* Patient characteristics associated with the successful transition to virtual care: lessons learned from the first million patients. *J Telemed Telecare*. 2023;29(8):621-631.
47. McAlarnen LA, Tsaih SW, Aliani R, Simske NM, Hopp EE. Virtual visits among gynecologic oncology patients during the COVID-19 pandemic are accessible across the social vulnerability spectrum. *Gynecol Oncol*. 2021;162(1):4-11.
48. Araújo LMB, Porto MV, Netto EM, Ursich MJ. Association of acanthosis Nigricans with race and metabolic disturbances in obese women. *Braz J Med Biol Res*. 2002;35(1):59-64.
49. Shivaprakash G, Basu A, Ashwin K, *et al.* Acanthosis Nigricansin PCOS patients and its relation with type 2 diabetes Mellitus and body mass at a tertiary care hospital in Southern India. *J Clin Diagn Res*. 2013;7(2):317-319.
50. Keen MA, Shah IH, Sheikh G. Cutaneous manifestations of polycystic ovary syndrome: a cross-sectional clinical study. *Indian Dermatol Online J*. 2017;8(2):104-110.
51. Dunaif A, Sorbara L, Delson R, Green G. Ethnicity and polycystic ovary syndrome are associated with independent and additive decreases in insulin action in Caribbean-Hispanic women. *Diabetes*. 1993;42(10):1462-1468.
52. Kauffman RP, Baker VM, DiMarino P, Gimpel T, Castracane VD. Polycystic ovarian syndrome and insulin resistance in white and Mexican American women: a comparison of two distinct populations. *Am J Obstet Gynecol*. 2002;187(5):1362-1369.
53. Engmann L, Jin S, Sun F, *et al.* Racial and ethnic differences in the polycystic ovary syndrome metabolic phenotype. *Am J Obstet Gynecol*. 2017;216(5):493.e1-493.e13.
54. Ezeh U, Ida Chen YD, Azziz R. Racial and ethnic differences in the metabolic response of polycystic ovary syndrome. *Clin Endocrinol (Oxf)*. 2020;93(2):163-172.
55. Ehrmann DA, Kasza K, Azziz R, Legro RS, Ghazzi MN. Effects of race and family history of type 2 diabetes on metabolic status of women with polycystic ovary syndrome. *J Clin Endocrinol Metab*. 2005;90(1):66-71.
56. Hillman JK, Johnson LNC, Limaye M, Feldman RA, Sammel M, Dokras A. Black women with polycystic ovary syndrome (PCOS) have increased risk for metabolic syndrome and cardiovascular disease compared with white women without PCOS. *Fertil Steril*. 2014;101(2):530-535.
57. Kumar A, Woods KS, Bartolucci AA, Azziz R. Prevalence of adrenal androgen excess in patients with the polycystic ovary syndrome (PCOS). *Clin Endocrinol (Oxf)*. 2005;62(6):644-649.
58. Welt CK, Arason G, Gudmundsson JA, *et al.* Defining constant *Versus* variable phenotypic features of women with polycystic ovary syndrome using different ethnic groups and populations. *J Clin Endocrinol Metab*. 2006;91(11):4361-4368.
59. Coney P, Ladson G, Sweet S, Legro R. Does polycystic ovary syndrome increase the disparity in metabolic syndrome and cardiovascular-related health for African-American women? *Semin Reprod Med*. 2008;26(1):035-038.
60. Harton J, Mitra N, Hubbard RA. Informative presence bias in analyses of electronic health records-derived data: a cautionary note. *J Am Med Inform Assoc*. 2022;29(7):1191-1199.
61. Boston Medical Center. *About BMC*. Accessed October 17, 2023. https://www.bmc.org/about-bmc
62. America's Most Socially Responsible Hospitals. *Lown Institute Hospital Index*. Accessed October 17, 2024. https://lownhospitalsindex.org/lists/2022-social-responsibility/
63. Bellavia A, Zota AR, Valeri L, James-Todd T. Multiple mediators approach to study environmental chemicals as determinants of health disparities. *Environ Epidemiol*. 2018;2(2):e015.
64. American College of Obstetricians and Gynecologists. *Racism in Obstetrics and Gynecology*. 2022. Accessed October 17, 2024. https://www.acog.org/clinical-information/policy-and-position-statements/statements-of-policy/2022/racism-in-obstetrics-gynecology